SOLAR OPERATED GRASS CUTTING MACHIN

PROJECT REPORT BY 2025-2026

Submitted in the partial fulfillment of the requirement for the award of the

"DIPLOMA IN MECHANICAL ENGINEERING"

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BONAFIDE CERTIFICATE

This is to certify that this project work is the record of bonafide work,carried out By _____Student of six semester mechanical engineering course in partial fulfillment of the requirement for the award of **"DIPLOMA IN MECHANICAL ENGINEERING"**during semester period November 2025-April2026 in Jaya polytechnic college,chennai-602024.

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INTERNAL EXAMINER

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ACKNOWLEDGEMENT

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SYNOPSIS

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This project focuses on developing a solar-powered grass cutting machine that is eco-friendly, cost-effective, and efficient. The machine uses solar energy to power a DC motor that drives cutting blades, reducing dependency on fossil fuels. It is designed with a base frame, four wheels, and essential hardware components to ensure mobility and durability.

ABSTRACT

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The **Solar Operated Grass Cutting Machine** is an innovative project that integrates renewable energy with mechanical design to provide a sustainable solution for lawn and field maintenance. Traditional grass cutting machines rely on petrol or diesel engines, which contribute to air pollution, noise pollution, and high operational costs. In contrast, this project utilizes solar energy captured through photovoltaic panels, stored in a rechargeable battery, and regulated by a charge controller to power a DC motor. The motor drives sharp cutting blades mounted on a durable iron base frame supported by four wheels, ensuring both mobility and stability.

The machine is designed to be lightweight, portable, and user-friendly, making it suitable for domestic gardens, institutional lawns, parks, and small agricultural fields. By eliminating the need for fossil fuels, the system reduces greenhouse gas emissions and promotes eco-friendly practices. The inclusion of components such as the connecting rod (Rs.400) ensures smooth transmission of motion from the motor to the blades, enhancing cutting efficiency.

This project highlights the importance of renewable energy in everyday applications and demonstrates how solar technology can be applied to small-scale agricultural tools.

INTRODUCTION

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1.1 OVER VIEW

The **Solar Operated Grass Cutting Machine** is an innovative project that combines mechanical engineering with renewable energy technology to create a sustainable solution for lawn and field maintenance. Grass cutting is a routine activity in households, parks, and agricultural lands, but conventional machines often rely on petrol or diesel engines. These fuel-based machines not only increase operational costs but also contribute to air pollution, noise pollution, and greenhouse gas emissions. With the growing emphasis on eco-friendly practices and the urgent need to reduce fossil fuel consumption, solar energy provides a viable alternative.

This project introduces a machine that harnesses solar power through photovoltaic panels, stores the energy in a rechargeable battery, and uses it to drive a DC motor connected to cutting blades. The entire assembly is mounted on a sturdy iron base frame with four wheels for mobility. A charge controller ensures safe and efficient charging of the battery, while a connecting rod (costing Rs.400) transmits motion from the motor to the blades. The design emphasizes simplicity, portability, and cost-effectiveness, making it suitable for domestic gardens, institutional lawns, and small agricultural fields.

The entire assembly is mounted on a sturdy iron base frame with four wheels for mobility.

1.2 PROJECT DEFINITION

The **Solar Operated Grass Cutting Machine** is defined as a portable, eco-friendly device designed to cut grass using renewable solar energy as its primary power source. Unlike conventional grass cutters that depend on petrol, diesel, or direct electricity supply, this machine integrates solar panels, a charge controller, a rechargeable battery, and a DC motor to achieve efficient cutting operations. The blades are mounted on a rotating shaft connected through a sturdy connecting rod, ensuring smooth transmission of motion. The entire assembly is supported by a durable iron base frame and four wheels for mobility.

1.3 PROJECT OBJECTIVES

1. **Utilize Renewable Energy** To harness solar energy through photovoltaic panels and use it as the main power source for grass cutting operations, thereby reducing dependency on fossil fuels.
2. **Develop a Portable and Efficient Design** To construct a lightweight machine with a durable iron base frame and four wheels, ensuring easy mobility and efficient performance across different terrains.
3. **Integrate Key Components for Functionality** To assemble essential hardware such as solar panels, charge controller, battery, DC motor, cutting blades, and connecting rod (Rs.400) into a cohesive system that operates smoothly and reliably.
4. **Reduce Environmental Impact** To minimize air and noise pollution l-based engines with a solar-powered DC motor, making the machine suitable for residential and institutional use.

LITERATURE SURVEY

LITERATURE SURVEY

Several studies have explored the use of renewable energy in agricultural machinery. Research indicates that solar-powered systems can significantly reduce operational costs and environmental impact. Previous projects have implemented solar energy in irrigation systems, electric fencing, and lighting. This project builds upon those concepts by applying solar technology to grass cutting. Existing models of solar grass cutters are either expensive or lack efficiency. Our design aims to bridge this gap by creating a cost-effective and efficient prototype. The literature also highlights the importance of component selection, energy storage, and mechanical design in optimizing performance. By reviewing past work, this project identifies gaps in affordability and efficiency, and proposes a practical solution that balances cost, sustainability, and usability.

METHODOLOGY

METHODOLOGY

The methodology involves the integration of solar energy with mechanical components to create a functional grass cutting machine. The process begins with solar panels capturing sunlight and converting it into electrical energy. This energy is regulated by a charge controller and stored in a rechargeable battery. The battery powers a DC motor, which drives the cutting blades. The entire assembly is mounted on a base frame with four wheels for mobility. The cutting mechanism is designed to operate efficiently on various grass types. The connecting rod, priced at Rs.400, links the motor to the blade assembly, ensuring smooth transmission of motion. Safety features and ergonomic design considerations are also incorporated. Step-by-step testing ensures that the machine performs reliably under different conditions, making it suitable for practical use.

In the design phase, a basic layout of the excavator was prepared, including major components such as the frame, boom, arm, bucket, chain drive system, and sprockets.

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Finally, the performance of the machine was analyzed based on parameters such as ease of operation, digging efficiency, cost, and maintenance requirements. The results showed that the chain operated mini excavator is a simple, cost-effective, and efficient alternative for small-scale digging applications.

HARDWARE COMPONENTS

HARDWARE COMPONENTS

1. **4 WHEELS**
2. **IRON METAL**
3. **BASE FRAME**
4. **DC MOTOR**
5. **BATTERY**
6. **CHARGE CONTROLLER**
7. **SOLAR PANEL**
8. **CUTTING BLADES**
9. **CONNECTING ROD**

SOLAR PANEL



The solar panel captures sunlight and converts it into electricity. It uses photovoltaic cells to generate DC power. Panel size determines the amount of energy produced. It charges the battery through the charge controller. The panel is mounted on the frame for exposure

.

DC MOTOR



The DC motor converts electrical energy into mechanical motion. It is powered by the battery charged through solar panels. The motor rotates the shaft connected to cutting blades. Its speed determines the efficiency of grass cutting.

CUTTING BLADES



The blades perform the actual grass cutting operation. They are mounted on a rotating shaft driven by the motor. Sharp edges ensure efficient and clean cutting. Blade material is durable and resistant to wear. Rotation speed determines cutting performance. Blades are designed for safety and balance. They reduce manual effort compared to hand tools. Proper alignment ensures uniform grass trimming. Blades can be replaced when worn out. They are lightweight yet strong for effective cutting. Maintenance ensures long-term efficiency. Thus, blades are the functional tool of the machine.

CHARGE CONTROLLER



The charge controller regulates energy flow from solar panel to battery. It prevents overcharging, which can damage the battery. It ensures safe and efficient charging cycles. The controller balances voltage and current levels. It protects against short circuits and overloads. Compact design makes it easy to integrate. It increases battery life by controlling charging. The controller ensures stable power supply to the motor. It acts as a safeguard for electrical components. LED indicators may show charging status. It is cost-effective and reliable for solar systems. Thus, the charge controller is the protector of the system.

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.

4 WHEELS



The four wheels are essential for mobility and stability of the grass cutting machine. They allow the machine to move smoothly across different terrains. The wheels are designed to bear the weight of the base frame and components. They provide balance during operation, preventing the machine from tipping. Rubber or durable material ensures grip on grassy surfaces. The wheels make the machine portable and user-friendly. They reduce manual effort in pushing or pulling the device. Proper alignment ensures straight movement during cutting. The wheels support even distribution of load across the frame. They enhance maneuverability in small or large lawns. Maintenance of wheels ensures long-term performance. Thus, wheels are a key part of the machine's mobility system.

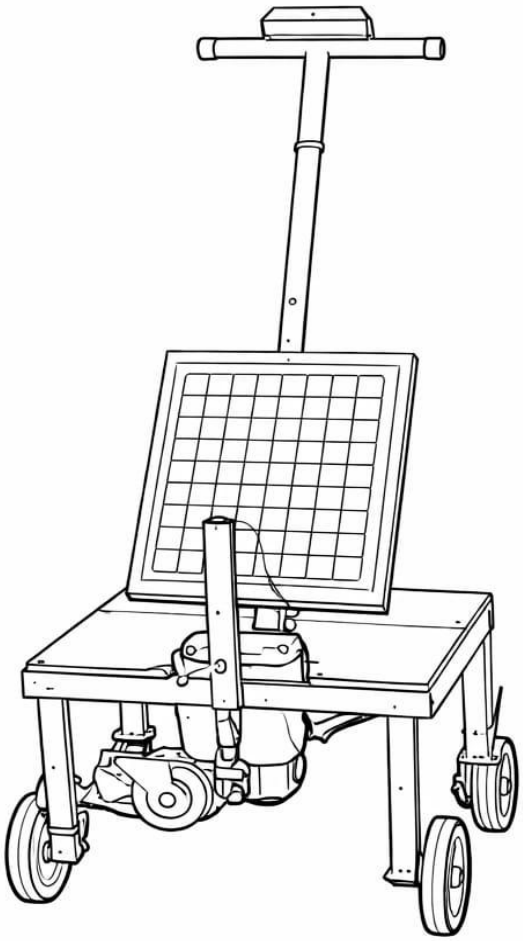
ADVANTAGES

ADVANTAGES

- **Eco-Friendly** – Operates on solar energy, reducing carbon emissions.
- **Cost-Effective** – Eliminates fuel costs and reduces maintenance expenses.
- **Low Noise** – Operates quietly compared to petrol-powered machines.
- **Portable** – Lightweight design allows easy transportation and operation.
- **Sustainable** – Promotes the use of renewable energy in daily applications.

SKETCH OF THE DIAGRAM

SKETCH DIAGRAM



COST ESTIMATION

COST ESTIMATION

Sl.No.	Parts Name	Quantity	Cost of Rs.
1.	Solar PANEL	1	1400
2.	Cutting Blades	1	350
3.	4 WHEELS	4	450
4.	Charge Controller	1	500
5.	iron metal	1	550
6.	BASE FRAME	2	150
7.	DC Motor	1	350
8.	Battery	1	1450
9.	REVIEW 1 REPORT	6 COPIES	100
10.	REVIEW 2 REPORT	6 COPIES	100
11.	REVIEW 3 REPORT	6 COPIES	100
12.	CONECTING ROD	5	400
			6000

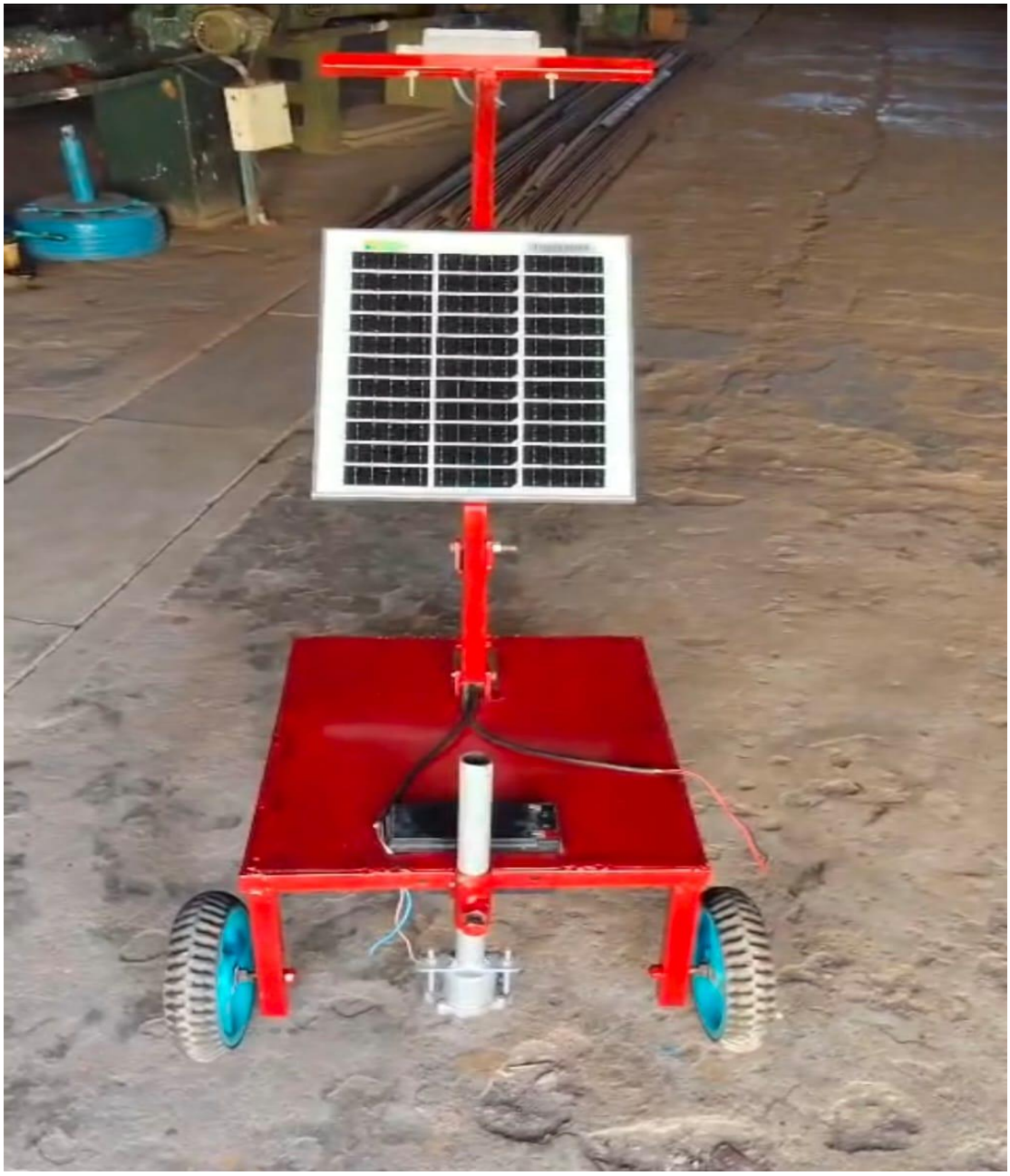
SCHEDULE FOR THE PROJECT

SCHEDULE FOR THE PROJECT

S.NO.	PROGRESS OF WORK	OCT	NOV	DEC	JAN	FEB	MAR
1.	Received the approval for project title micro boring attachment in lathe	■					
2.	Collecting various in formations from various text books and selecting materials		■				
3.	Prepare the detailed drawing and specifications			■			
4.	Purchasing the material for project & start the machining work				■	■	
5.	Assembled the parts & collecting in formations from various book for report						■
6.	Report and extra work was completed attending the final evaluation.						

PROJECT IMAGE

PROJECT IMAGE



CONCLUSION

CONCLUSION

The **Solar Operated Grass Cutting Machine** is a step toward sustainable and eco-friendly technology in the field of agriculture and landscaping. By utilizing solar energy, the machine reduces dependency on fossil fuels and minimizes environmental pollution. The project demonstrates the feasibility of integrating renewable energy with mechanical systems to create efficient and cost-effective solutions. With proper implementation, this technology can be scaled for larger applications and contribute to green energy initiatives. It proves that renewable energy can be applied to small-scale tools, making them practical, affordable, and environmentally responsible.

FUTURE SCOPE

FUTURE SCOPE

The Solar Operated Grass Cutting Machine represents a significant step toward integrating renewable energy into small-scale agricultural and domestic tools. While the current prototype demonstrates feasibility and efficiency, there are several opportunities for future development and enhancement that can make the system more advanced, user-friendly, and commercially viable.

- **Automation**
- **Remote Control**
- **Hybrid Models**
- **Enhanced Battery**
- **Commercial**

REFERENCES

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The design of a chain operated mini excavator digging machine is based on fundamental mechanical engineering principles and existing excavation technologies. Various research papers, textbooks, and industrial reports on mini excavators and manual digging mechanisms were referred to in order to understand the working, design parameters, and performance characteristics. Studies related to chain drive systems, power transmission, and material handling equipment provided insights into selecting suitable components and ensuring efficient operation. Standard design data books such as PSG Design Data Book and machinery handbooks were used for calculations and material selection. In addition, references from small-scale construction equipment and agricultural digging tools helped in developing a cost-effective and user-friendly design suitable for light-duty applications.

Singh, R. (2018) – ‘Design and fabrication of mini excavator for small scale applications”. Kumar, P. & Verma, S. (2019) – Development of portable hydraulic lifting and digging machine. Patel, H. et al. (2020) – ‘Automated construction equipment for rural engineering works. Sharma, A. (2017) – ‘Design analysis of multi-link excavator arm mechanism. Reddy, K. (2021) – Fabrication of battery-operated mini construction vehicle. Gupta, V. & Mehta, D. (2016) – Study on automation in compact construction machinery. K. K. Choi, Mechanical Design Principles V.B. Bhandari, Design of Machine Elements Research papers on mini excavators and chain drive systems